Modeling of Grid-Forming and Grid-Following Inverters for Dynamic Simulation of Large-Scale Distribution Systems

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***Abstract*—Historically, distribution system planning studies mainly focused on steady state and quasi-steady state analysis, with limited attention paid to dynamic analysis. This paper develops three-phase, electromechanical models for both grid-forming and grid-following inverters, and integrates them into an open source, three-phase distribution network solver, thereby enabling dynamic simulation of large-scale, three-phase unbalanced distribution systems with high penetration of inverter-based DERs. The proposed inverter models are validated against electromagnetic simulation and field test data from the CERTS/AEP microgrid testbed, and simulated in an islanded 5252 node distribution system in the GridLAB-D simulation environment. Simulation verifies the effectiveness of the proposed inverter models for large-scale distribution system analysis. Study results show that compared to traditional grid-following inverters, the high penetration of grid-forming inverters can improve the voltage and frequency stability of islanded distribution systems.**

***Index Terms*—Inverter, grid-forming, grid-following, dynamic simulation, modeling, distribution systems.**

I. Introduction

POWER systems are undergoing a major transition with increasing penetrations of distributed energy resources (DERs) connected at the distribution level, including both synchronous generator-based and inverter-based DERs [1], [2]. Historically, distribution systems seldom have had dynamic stability issues because of the existence of a strong substation voltage source and a limited number of DERs. However, the emerging generation of resilient distribution systems is expected to allow portions of, or even all the distribution feeder, to work in islanded modes when the substation voltage source is lost because of events like extreme weather and natural disasters [3]. For such islanded distribution systems with high penetration of DERs, dynamic stability becomes an operational concern.

Historically, distribution system planning and protection studies have mainly focused on steady state and quasi-steady state analysis, with limited attention paid to dynamic analysis [4]. The work by [5] proposed a three-phase dynamic model of synchronous generators for dynamic simulation of distribution systems, but inverter models are still lacking for distribution system studies. Existing dynamic models of inverters are mostly electromagnetic models and sequential electromechanical models, both of which have limited utility for distribution system analysis. Electromagnetic models can be used to study the performance of a single inverter, but their use in simulating large-scale distribution systems, which typically have thousands of nodes in North America [6], is not practical. Sequential electromechanical models are typically used for transmission system studies. The work by [7]–[9] developed different positive-sequence models of inverters and investigate their impact on the dynamic stability of transmission systems. The work by [10] developed negative-sequence models of inverters that can be used to study the unsymmetrical faults of transmission systems. However, the sequential electromechanical models are mainly used for transmission systems where the network is always assumed to be three-phase balanced except for the fault location, and hence a single-phase network solver would be sufficient. In contrast, distribution networks are usually three-phase unbalanced in normal operations because of their three-phase unbalanced construction with single- and double-phase laterals [4], [6], where the single-phase network solver is not suitable and three-phase network solvers are usually required. Therefore, it is not practical to link sequential electromechanical models of inverters to distribution network solvers.

From the controller design perspective, inverter controls can be classified as one of two basic types: grid-following and grid-forming [1]. Grid-following control is widely used for grid-connected inverters. It makes the inverter behave approximately like a current source. In recent years, studies have shown that as the penetration of grid-following inverters increases in

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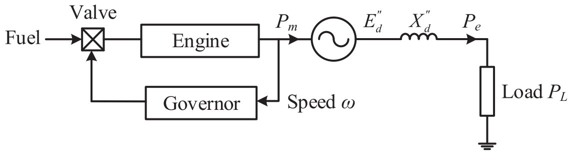

Fig. 1. Generator supplying isolated load.

transmission systems, with a corresponding decrease in synchronous generators, the dynamic stability is harmed [11], [12]. In contrast, grid-forming control is considered an emerging technology for power systems [13]–[15] that typically makes the inverter behave like a voltage source. Different grid-forming controls have been proposed in recent years [16]–[18], and their capabilities to maintain system stability have been examined in small-scale systems such as microgrids [19], [20]. In addition, simulation studies performed at transmission levels indicate that grid-forming inverters can improve the voltage and frequency stability of bulk power systems [8], [9], [12], [13], [21]. However, the impacts of grid-following and grid-forming inverters on large-scale distribution systems have not been thoroughly investigated because of the lack of appropriate models.

This paper develops three-phase, electromechanical models for both grid-following and grid-forming inverters, and integrates them into an open source, three-phase distribution network solver, thereby enabling dynamic simulation of large-scale, three-phase unbalanced distribution systems that feature high penetration of inverter-based DERs. The inverter models are validated against electromagnetic transient simulation and field test data from the Consortium for Electric Reliability Technology Solutions (CERTS)/American Electric Power (AEP) microgrid testbed [22], and evaluated in an islanded, 5252 node distribution feeder that has multiple synchronous generator-based and inverter-based DERs. Simulation verifies the effectiveness of the proposed inverter models for large-scale distribution system analysis. Study results show that, compared to traditional grid-following inverters, the high penetration of grid-forming inverters can improve the voltage and frequency stability of islanded distribution systems.

II. Inertia of Synchronous Generators

A synchronous generator behaves approximately like a voltage source behind its subtransient reactance X_d'' during load transients, as shown in Fig. 1 [23]. During a load step, the generator draws necessary currents to meet any load changes, while its internal voltage E_d'' remains constant. However, the prime mover of a synchronous generator usually responds slowly [23]. This results in an imbalance between the generator's electrical power P_e and mechanical power P_m, forcing the rotor speed to change, as shown in Fig. 2. Equation (1) describes the swing equation of a synchronous generator, where H is the inertia constant and ω is the rotor speed. A larger H results in a smaller rate of speed change.

$$2H\frac{d\omega}{dt} = P_m - P_e \tag{1}$$

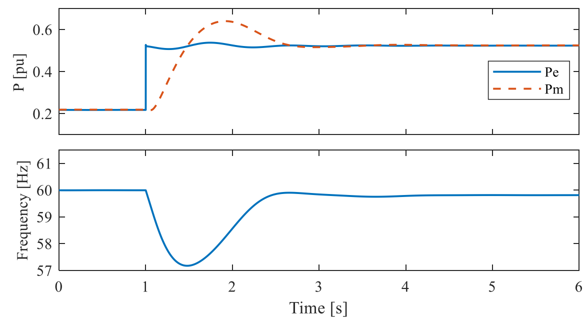

Fig. 2. Response of a synchronous generator to a load step.

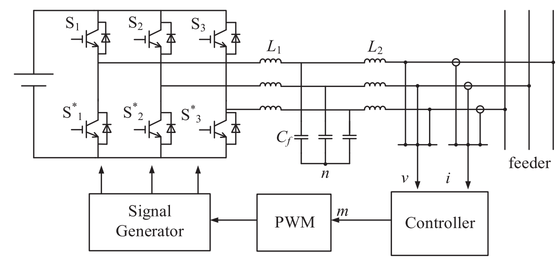

Fig. 3. A typical two-level, three-phase voltage-sourced inverter.

Although the inertia constant H is important to maintaining the frequency stability of a synchronous generator, a more fundamental reason for the frequency change is the imbalance between P_m and P_e during load transients caused by the slow response of P_m.

In a highly inverter-penetrated distribution system, if the inverter-based DERs could quickly respond to any load disturbances, the imbalance between P_m and P_e of synchronous generators could be reduced, helping to improve the frequency stability. The following sections introduce two typical inverter controls and discuss their different responses to load disturbances.

III. Inverter Controls

Most inverter-based DERs use the two-level, three-phase voltage-sourced inverter, as shown in Fig. 3 [24]. Although the device is called a "voltage-sourced" inverter, different control strategies can cause different dynamic behaviors of inverters. This section introduces two basic types of controls, known as "grid-following" and "grid-forming."

A. Grid-Following Concept

Currently, most grid-connected, inverter-based DERs use grid-following control, which typically uses a phase-lock-loop (PLL) and a current control loop to achieve fast control of the inverter's output currents [24]. Grid-following control makes the voltage-sourced inverter behave approximately like a current source, as shown in Fig. 4(a). The advantage of this control is that the currents can be quickly regulated. However, because grid-following control does not control the voltage and frequency, it relies on an external voltage source to provide the

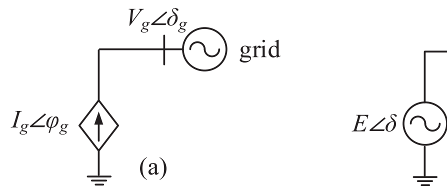

Fig. 4. Basic concepts of inverter controls: (a) grid-following and (b) grid-forming.

voltage and frequency references. During load disturbances, grid-following inverters maintain their output currents or output power approximately constant.

B. Grid-Forming Concept

In contrast, grid-forming control controls the voltage and frequency of the inverter, making the voltage-sourced inverter behave approximately like a voltage source, as shown in Fig. 4(b). Because the voltage and frequency remain constant, the grid-forming inverters can work in the stand-alone mode and track the loads [25]. To achieve parallel operation of multiple grid-forming inverters, different control strategies have been proposed, including droop control [16], [25], virtual oscillator control [18], virtual synchronous machines [17], etc.

IV. Modeling of Inverters for Dynamic Simulation of Large-Scale Distribution Systems

To investigate the impact of grid-following and grid-forming inverters on the voltage and frequency stability of distribution systems, appropriate dynamic models need to be developed. This section develops three-phase, electromechanical models for both grid-following and grid-forming inverters, and their interfaces to the three-phase distribution network power flow solver.

A. Modeling and Simulation Assumption

The following assumptions are made before developing the inverter models.

First, the DC bus voltage of inverters is assumed to be constant. This assumption applies to DERs such as energy storage systems with the state-of-charge in normal ranges, and natural-gas-driven inverters with a surge module connected at the DC bus [26], but does not apply to DERs such as single-stage photovoltaic (PV) inverters. The work reported by [27] pointed out that the DC bus voltage of grid-forming-controlled PV inverters might collapse during overload events. The DC bus voltage collapse phenomenon is out of the scope of this paper and will be studied in future work.

Second, the inverter models are developed for examining the system dynamic stability under relatively small disturbances, such as a trip of generators and lines, and step change in loads. The models should not be used for examining the transient stability under large disturbances such as short-circuit faults. Inverter-based DERs may respond to faults in different ways. According to IEEE std. 1547-2018 [28], the grid-following inverters might cease to energize or trip off during faults. For grid-forming inverters, the field test by [22] showed that a 60 kW commercially available grid-forming inverter can provide 6 pu fault current in a high-impedance fault, while the test results reported by [29] indicated that a 5 kW commercially available grid-forming inverter quickly turned off the switches to prevent the current from exceeding 1.5 pu. Considering the various responses of inverter-based DERs to faults, fault ride-through controls are not modeled in this paper and will be studied in future work.

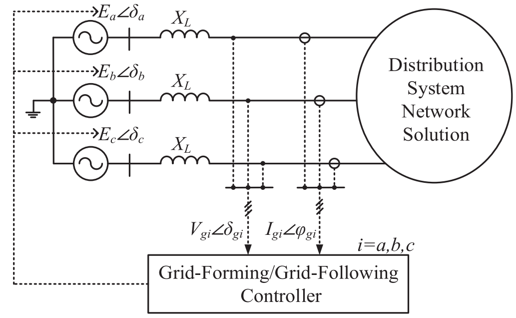

Fig. 5. Inverter equivalent circuit and controller.

B. Inverter Model

By assuming the DC bus voltage to be constant, the inverter main circuit in Fig. 3 can be modeled as a three-phase controllable voltage source behind the coupling reactance X_L, as shown in Fig. 5. The X_L accounts for the inverter's filter inductances L_1 and L_2. In this paper, the value of X_L is set as 0.1 pu on an inverter rating base. Filter capacitance C_f is ignored due to its small value in the fundamental frequency. As shown in Fig. 5, the magnitude E_i (i = a,b,c, also applies to other variables in the paper) and phase angle δ_i of the inverter internal voltage $E_i\angle\delta_i$ are regulated by the inverter controller, which could be either grid-forming or grid-following. The inputs of the inverter controller are the three-phase terminal voltages $V_{gi}\angle\delta_{gi}$ and currents $I_{gi}\angle\varphi_{gi}$.

Based on the above analysis, two assumptions underlie the inverter model being an electromechanical model rather than an electromagnetic model. First, a phasor representation is used for the voltage source, and second, the dynamic of the inverter filter is neglected, because the inverter filter is described by an algebraic term "jX_L" instead of a differential term. Given these two assumptions, it is possible to link the inverter model to a distribution network power flow solver that models the network using algebraic equations. Therefore, simulation efficiency can be significantly improved compared to the electromagnetic simulation that models the whole network using differential equations.

To be linked with a three-phase distribution network power flow solver, the voltage source model is converted to its Norton equivalence, as shown in Fig. 6(a) and (b). The three-phase internal current $I_{Ii}\angle\varphi_{Ii}$ and the admittance Y_L are calculated using (2) and (3).

$$I_{Ii}\angle\varphi_{Ii} = \frac{E_i\angle\delta_i}{jX_L}, i = a, b, c \tag{2}$$

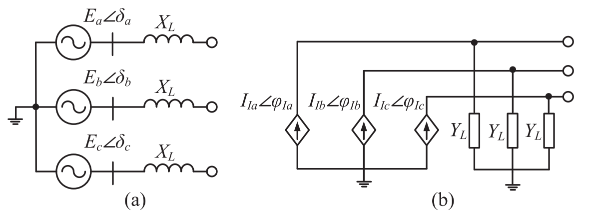

Fig. 6. Inverter equivalent circuits: (a) Inverter internal voltage source and (b) Norton equivalent current source.

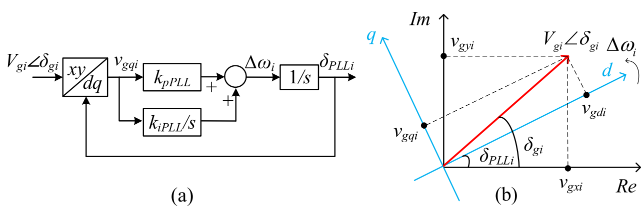

Fig. 7. PLL: (a) control block and (b) *xy* and *dq* frame coordinate systems.

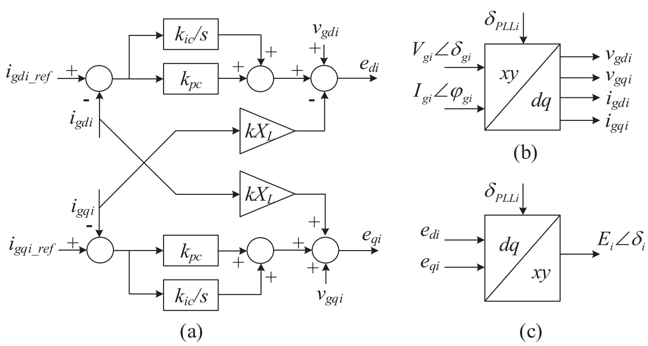

Fig. 8. Control block of the current control loop: (a) current loop, and (b) and (c) coordinate transformations.

$$Y_L = \frac{1}{X_L} \tag{3}$$

C. Grid-Following Control Dynamic Model

In most commercially available dynamic stability simulation tools, grid-following inverters are modeled as controllable current sources, and the inner control loops are ignored [30], [31]. This paper instead seeks to faithfully represent the real control strategies used by grid-following inverters. Therefore, two key components of grid-following control—PLL and current control loop, are modeled. Figs. 7 and 8 show the control blocks of a typical synchronous reference frame PLL and a typical current control loop, respectively. The control objective of the PLL is to estimate the phase angle of the grid voltage. As shown in Fig. 7(a), the PLL uses a proportional-integral (PI) controller to control the component of the grid voltage on q-axis, v_{gqi}, to be zero by increasing or decreasing the virtual angular frequency $\Delta\omega_i$. The integration of $\Delta\omega_i$ is the estimated phase angle δ_{PLLi}. δ_{PLLi} should equal δ_{gi} in the steady state. Fig. 7(b) illustrates how the PLL tracks the phase angle of the grid voltage. k_{pPLL} and k_{iPLL} are the proportional and integral gains of the controller.

The current control loop quickly regulates the active and reactive currents injected into the grid by adjusting the inverter internal voltage $E_i\angle\delta_i$ rapidly. In this study, the output current of each phase is independently controlled in the *dq* frame through the PI controller shown in Fig. 8(a), where k_{pc} and k_{ic} are the controller parameters. The grid voltage $V_{gi}\angle\delta_{gi}$ and current $I_{gi}\angle\varphi_{gi}$ are transferred from the *xy* frame to the *dq* frame based on the phase angle δ_{PLLi} obtained from the PLL, as shown in Fig. 8(b). The *xy* frame here refers to the frame that is used to perform the power flow analysis of the distribution network. The outputs of the current loop are the inverter internal voltages in the *dq* frame, e_{di} and e_{qi}. They are then transferred to the *xy* frame, as shown in Fig. 8(c). Once $E_i\angle\delta_i$ is obtained, (2) and (3) can be used to get the Norton equivalent circuit, then the inverter model can be linked to the three-phase network solution.

When connected to an unbalanced distribution network, the grid-following inverters could be controlled to inject a combination of positive-, negative-, and possibly zero-sequence currents to the grid. This can be done by appropriately setting the current reference $I_{gi\_ref}\angle\varphi_{gi\_ref}$ for each phase. In this study, grid-following inverters are controlled to only inject positive-sequence current into distribution systems. With this assumption made, the current reference for each phase, $I_{gi\_ref}\angle\varphi_{gi\_ref}$, can be obtained using (4) and (5). $V_{g1}\angle\delta_{g1}$ is the positive-sequence voltage, which can be extracted from the three-phase inverter terminal voltage $V_{gi}\angle\delta_{gi}$. $I_{g1\_ref}\angle\varphi_{g1\_ref}$ is the positive-sequence current reference of the inverter, P_{ref} and Q_{ref} are the references of active and reactive power, and $\alpha = e^{j2\pi/3}$. Once the current reference for each phase $I_{gi\_ref}\angle\varphi_{gi\_ref}$ is obtained using (5), it is then transferred to the *dq* frame to obtain the references i_{gdi_ref} and i_{gqi_ref} that serve as the inputs of the inner current loop, as shown in Fig. 8(a).

$$I_{g1\_ref}\angle\varphi_{g1\_ref} = \left(\frac{P_{ref} + jQ_{ref}}{V_{g1}\angle\delta_{g1}}\right)^* \tag{4}$$

$$\begin{bmatrix} I_{ga\_ref}\angle\varphi_{ga\_ref} \\ I_{gb\_ref}\angle\varphi_{gb\_ref} \\ I_{gc\_ref}\angle\varphi_{gc\_ref} \end{bmatrix} = \begin{bmatrix} 1 \\ \alpha^2 \\ \alpha \end{bmatrix} I_{g1\_ref}\angle\varphi_{g1\_ref} \tag{5}$$

Note that in recent years, different auxiliary controls have been proposed for grid-following inverters, aiming to provide voltage and frequency support to power systems, such as volt-var, frequency-watt, synthetic inertia, etc. [32], [33]. These controllers are designed as outer control loops that modify P_{ref} and Q_{ref}. The outer control loops are subject to the bandwidth of the inner control loops, so their responses are usually slow. The analysis and experimental results reported by [33] show that a first-order low-pass filter with a time constant of 1 s is needed for synthetic inertia control to guarantee local stability. The work reported by [8] indicates the slow response of frequency-watt control leads to a poor frequency response of bulk power systems. The grid support functions of grid-following inverters are not considered in this paper. Instead, this study focuses on modeling detailed inner control loops for grid-following inverters; future work will model different outer controls.

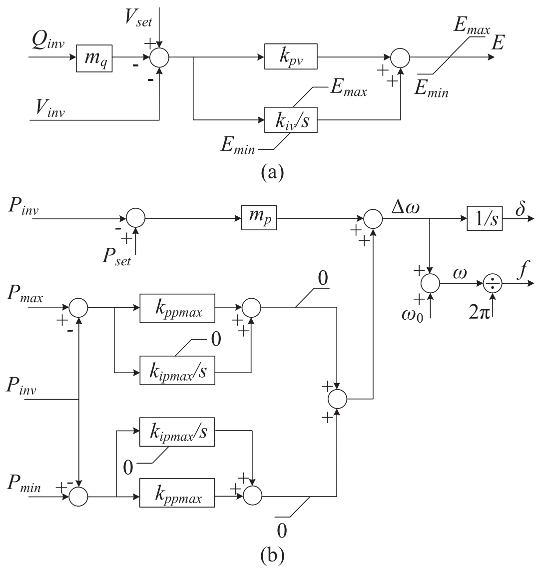

Fig. 9. CERTS Droop Control: (a) Q-V droop control and (b) P-f droop control and overload mitigation control.

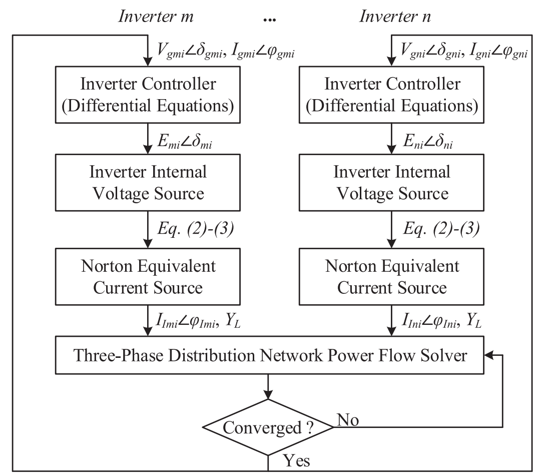

Fig. 10. Interaction between inverter models and the power flow solver.

D. Grid-Forming Control Dynamic Model

As discussed in Section III-B, different grid-forming controls have been proposed in recent years. In this study, CERTS droop control was selected, because extensive field tests have been conducted at the CERTS/AEP microgrid test bed during the past decade [22], so it can be considered a relatively mature technology. As shown in Fig. 9(a) and (b), CERTS droop control controls the voltage magnitude *E* and frequency *f* of the inverter internal voltage $E\angle\delta$ according to the *Q-V* droop control and *P-f* droop control, respectively. P_{set} and V_{set} are the set points of active power and voltage, and m_q and m_p are the droop gains. The *Q-V* droop control mitigates circulating reactive power between paralleled grid-forming inverters, and the *P-f* droop control synchronizes the phase angles of inverters and ensures power sharing between them. In addition, an overload mitigation control [19] is added on the *P-f* droop control to (1) prevent the output power of inverters from exceeding the maximum $P_{\max}$ or dropping below the minimum $P_{\min}$ during partial overload events in a microgrid, and (2) help to activate under-frequency load shedding when all inverters in a microgrid are overloaded. k_{pv} and k_{iv} are the gains of the voltage control loop. k_{ppmax} and k_{ipmax} are the gains of the overload mitigation control, and ω_0 is the rated angular frequency.

To link the controller to a three-phase distribution network power flow solver, the inputs of the controller, P_{inv}, Q_{inv}, and V_{inv}, are calculated using (6)–(8). P_{inv} and Q_{inv} refer to the sum of the inverter three-phase active and reactive power, respectively, and V_{inv} is the average of the magnitude of the inverter three-phase terminal voltages. A first-order low-pass filter with a time constant $T = 0.01$ s is added in (6)–(8), which is in accordance with the electromagnetic model [34]. As Seen in Fig. 9(a), the *Q-V* droop control only controls the average value of the inverter terminal voltages, and the inverter three-phase terminal voltages are unbalanced because of the unbalanced structure of the distribution network.

$$P_{inv} = \frac{1}{1+Ts}(P_a + P_b + P_c) \tag{6}$$

$$Q_{inv} = \frac{1}{1+Ts}(Q_a + Q_b + Q_c) \tag{7}$$

$$V_{inv} = \frac{1}{3(1+Ts)}(V_{ga} + V_{gb} + V_{gc}) \tag{8}$$

Although the inverter terminal voltages are unbalanced, the controller keeps the inverter internal voltages to be always three-phase balanced, as shown in (9)–(10). Keeping the inverter internal voltages to be balanced would be beneficial for mitigating the overall voltage imbalance of the entire distribution network when more and more such inverters are connected to the system. Note that in this way the inverter output currents are no longer balanced because of the unbalanced structure of the distribution system.

$$E_a = E_b = E_c = E \tag{9}$$

$$\begin{cases} \delta_a = \delta \\ \delta_b = \delta - \frac{2}{3}\pi \\ \delta_c = \delta + \frac{2}{3}\pi \end{cases} \tag{10}$$

V. Interaction Between Inverter Models and the Distribution Network Solver

The open-source software GridLAB-D was selected as the three-phase distribution network power flow solver [35]. The network and loads in GridLAB-D are modeled using a per phase representation, allowing a complete representation of line and load imbalances. The power flow solution used in GridLAB-D is an extension of the current injection method described in [36].

Fig. 10 shows the interaction between inverter models and the network solver. All inverter models have the same simulation time step. As seen in the figure, the inverters interact with each other through the three-phase distribution network power flow solver. With this modeling approach, the developed inverter models can also be linked to other three-phase distribution

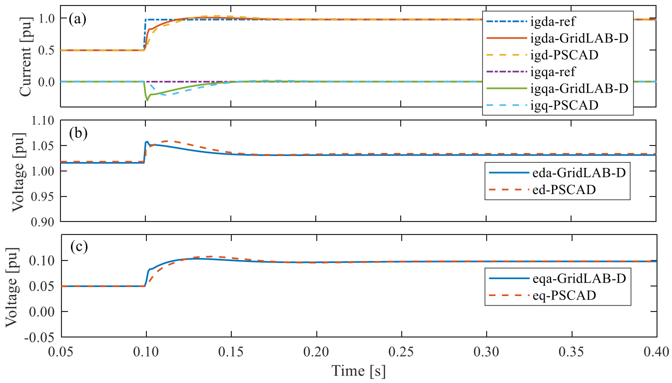

Fig. 11. GridLAB-D and PSCAD simulation of a grid-following inverter.

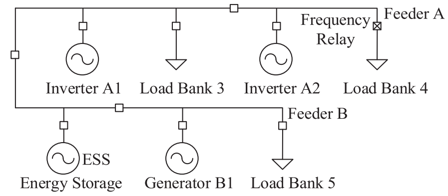

Fig. 12. One-line diagram of the reduced CERTS/AEP microgrid.

network power flow solvers. More detail about how the dynamic simulation is setup can be found in [5], [35].

VI. Model Validation in Small Systems

To verify the developed inverter models, the three-phase, electromechanical simulation results from GridLAB-D are compared to the electromagnetic simulation results from PSCAD [37]. Moreover, field test results from the CERTS/AEP microgrid are used to validate the grid-forming inverter model. All the parameters of inverters and networks remain the same in both the PSCAD and GridLAB-D simulation.

A. Grid-Following Inverter

Because grid-following control aims to track P_{ref} and Q_{ref}, a step change in P_{ref} is used to validate the proposed grid-following inverter model. In this simulation, a 100 kW grid-following inverter is connected to an infinite bus through a short distribution line of 0.0217 + j0.0223 pu, and P_{ref} is changed from 0.5 pu to 1 pu at 0.1 s. Fig. 11(a)–(c) shows GridLAB-D and PSCAD simulation results. As seen in Fig. 11(a), the inverter output currents i_{gda} and i_{gqa} respond quickly to the change in current references and reach a steady state within 0.1 s. Fig. 11(b) and (c) show the inverter internal voltages e_{da} and e_{qa}.

The simulation results from GridLAB-D are slightly different than the simulation results from PSCAD. This is because in electromagnetic simulation, the fast response current control loop interacts with the dynamics of the inverter filter inductance and line impedance, but those dynamics are ignored in the electromechanical simulation, resulting in the difference in the simulation results.

B. Grid-Forming Inverter

Given the fact that both *P-f* droop control and overload mitigation control are activated during the load shedding process, two under-frequency load shedding field test results from the CERTS/AEP microgrid were used to validate the proposed grid-forming inverter model. In the first test, under-frequency load shedding is performed in a purely grid-forming inverter-based microgrid. In the second test, under-frequency load shedding is performed in a grid-forming inverter and a synchronous generator mixed microgrid. Fig. 12 shows the one-line diagram of the reduced CERTS/AEP microgrid, and Fig. 13(a) and (c) present the two field test results.

In the first test, grid-forming inverters A1 (100 kW), A2 (60 kW) and energy storage system (ESS, 100 kW) work together in islanded mode, and the tripping of the ESS results in the output power of A1 and A2 exceeding their maximum power outputs. Therefore, their overload mitigation controllers are activated to reduce the frequency rapidly. The frequency relay installed at Load Bank 4 detects the under-frequency event and trips Load Bank 4, resulting in the survival of the microgrid. Fig. 13(b) shows both GridLAB-D and PSCAD simulation results. The fA1 and fA2 in Fig. 13(b) are obtained from the inverter controllers, as shown in Fig. 9(b). It can be seen that their values are different during load transients, indicating inverters interact with each other, and the GridLAB-D and PSCAD simulation match very well, even though they use different modeling approaches. The simulation results are also similar to the field test results. The second test is similar to the first, but A2 is replaced by synchronous generator B1 (93 kW). Fig. 13(c) and (d) show that the GridLAB-D simulation, PSCAD simulation, and field test results match well with each other. The slight difference between GridLAB-D and PSCAD simulation can be attributed to the difference in the synchronous generator models used in GridLAB-D and PSCAD. Note that in Fig. 13(d) fA1 is obtained from the inverter controller, and fB1 refers to the speed of the synchronous generator.

Both load shedding tests have verified the proposed grid-forming inverter models. Detailed explanations and parameters of the two tests are provided in [19].

VII. Simulation in a Large-Scale, Three-Phase Unbalanced Distribution System

To evaluate the feasibility of the developed inverter models for dynamic stability simulation of large-scale, three-phase unbalanced distribution systems, the prototypical distribution feeder, R3-12.47-3, developed at Pacific Northwest National Laboratory was selected as the studied system [38]. The feeder has 5252 nodes, including 2002 nodes at the primary voltage side and 3250 nodes at the secondary voltage side, representing a typical three-phase unbalanced distribution feeder in a heavily populated suburban area in North America. The rated voltage on the primary side is 12.47 kV, and the rated voltages on the secondary side are 480 V and 120 V. The total load is around 8 MW and has a power factor of 0.9.

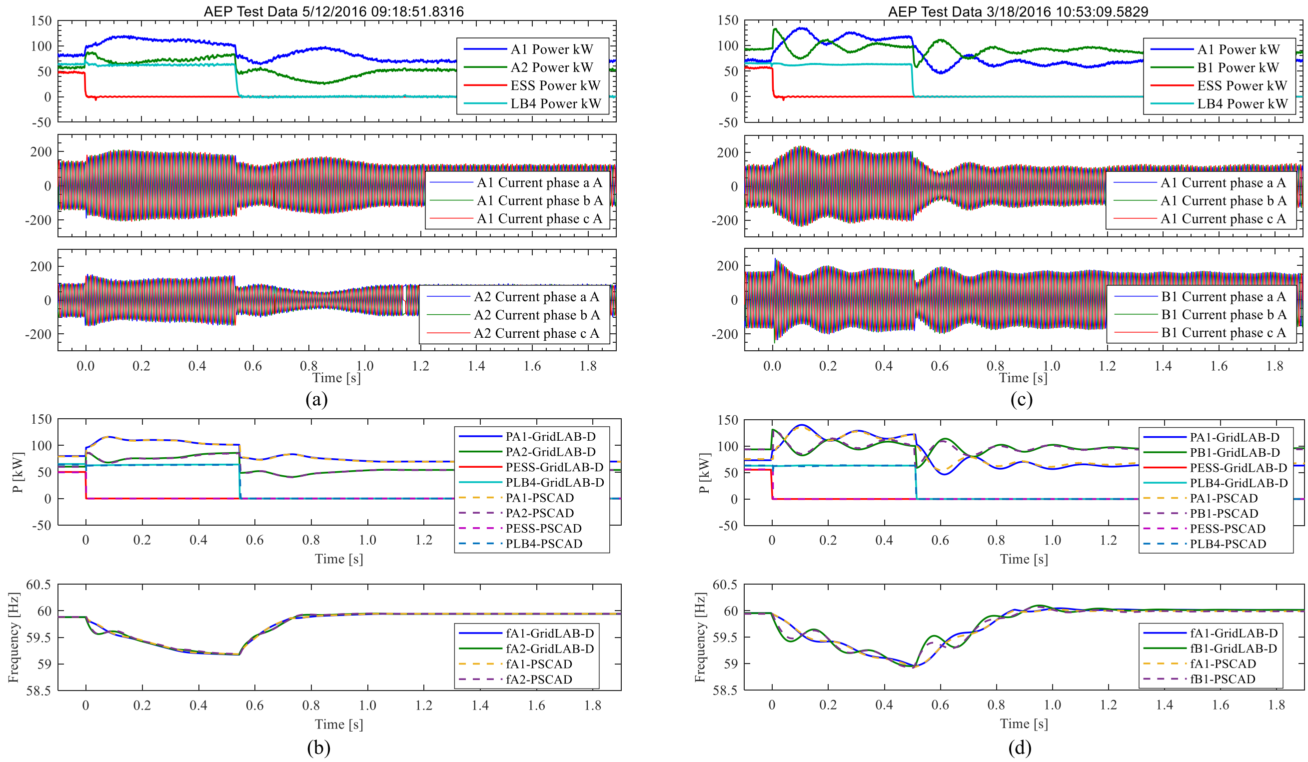

Fig. 13. Field test and simulation results of two under-frequency load shedding events from the CERTS/AEP microgrid. (a) and (b): Load shedding in a purely inverter-based microgrid. (c) and (d): Load shedding in an inverter and generator mixed microgrid.

The following modifications are made to conduct the dynamic stability simulation. (1) The substation voltage source is disconnected to examine the capability of the DERs to maintain the stability of the islanded distribution system. (2) Three 2 MW diesel generators, each having an H of 1 s, are installed in the feeder. The generator model is based on the work in [5], but additional Q-V droop control is modeled to mitigate circulating reactive power during parallel operations. (3) Ten 500 kW utility-scale inverter-based DERs are installed in the feeder. The inverters can be either grid-following or grid-forming controlled, and the controller parameters are provided in Tables II and III of the Appendix. (4) Approximately 25% of the loads are equipped with under-frequency load shedding devices that have a load shedding frequency of 59 Hz and a time delay of 0.05 s [39]. (5) The switching actions of regulators and shunt capacitors are ignored considering their slow responses.

Fig. 14 shows the studied distribution system and the locations of the DERs. The initial condition is that the three synchronous generators are each dispatched to output around 1.8 MW, and the 10 inverters are categorized as three groups: Inverters 1–3 each output 100 kW, Inverters 4–8 each output 300 kW, and Inverters 9 and 10 each output 500 kW. Note that Inverters 9 and 10 are dispatched at their maximum power outputs. The voltage set points V_{set} of generators and grid-forming inverters are set at 1 pu. The grid-following inverters generate power at unity power factor. Three cases are simulated as follows. In Case A, Generator 1 is tripped, and all inverters are grid-following controlled. In Case B, Generator 1 is tripped, and all inverters are grid-forming controlled. In Case C, Generator 2 is tripped after Case B occurs.

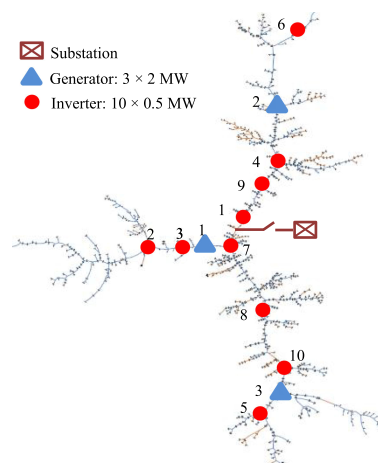

Fig. 14. The studied 5252 node islanded distribution system.

A. Loss of One Generator With All Grid-Following Inverters

Figs. 15 and 16 show the simulation results for Case A. The tripping of Generator 1 results in a significant frequency oscillation in the islanded distribution system, triggering under-frequency load shedding.

In Fig. 15(a), the frequencies of generators refer to their speeds, and the frequencies of inverters are measured by their

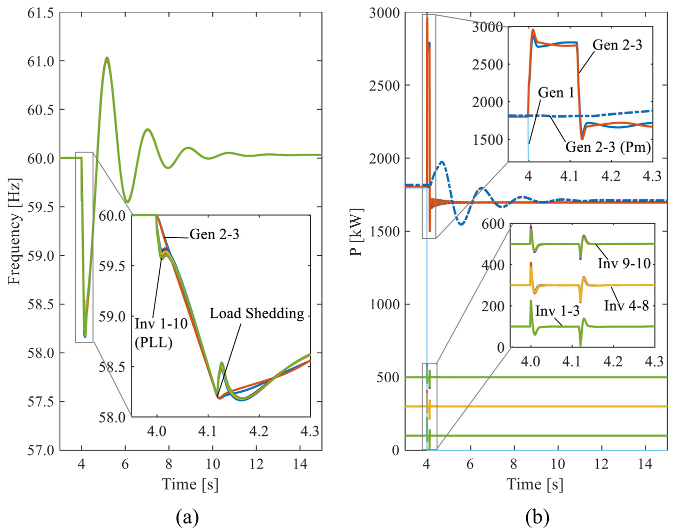

Fig. 15. Case A, grid-following: (a) frequency and (b) active power.

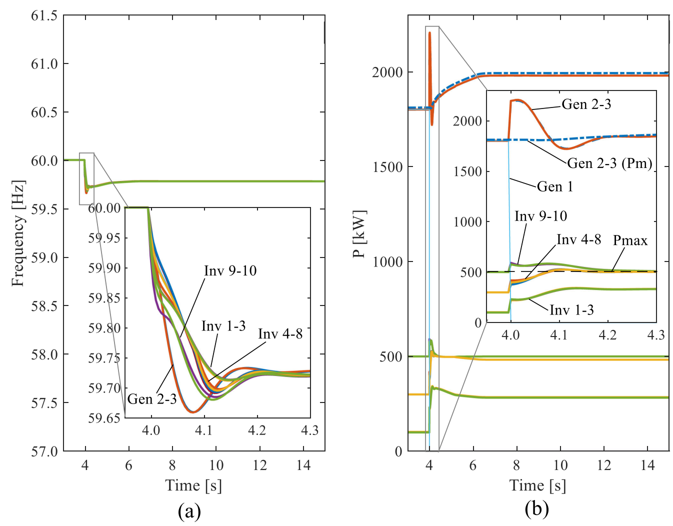

Fig. 17. Case B, grid-forming: (a) frequency and (b) active power.

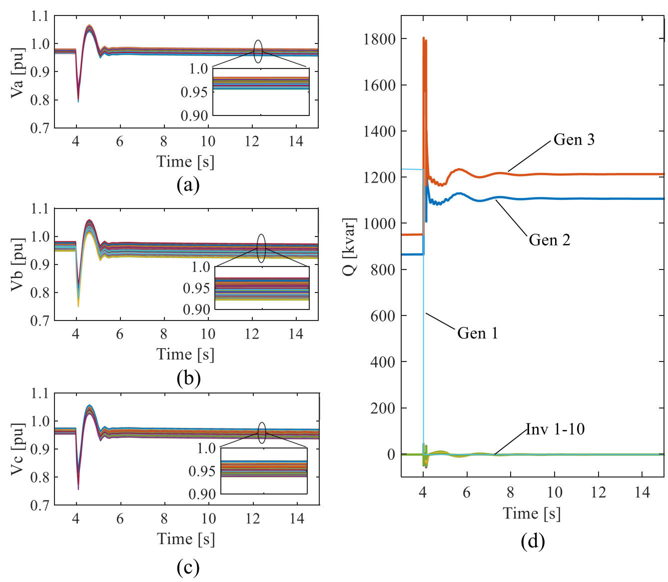

Fig. 16. Case A, grid-following: (a)–(c) voltages and (d) reactive power.

PLLs. As shown in Fig. 15(b), at the beginning of the event both generators and inverters increase their output power to meet the loss of generation. This is because both generators and inverters behave as voltage sources behind reactances, and their internal voltages are constant initially. However, the grid-following controllers quickly regulate the output power of the inverters back to their power references by changing the inverter internal voltages rapidly, as explained in Section IV-C. Therefore, Generators 2 and 3 have to take all the output power of Generator 1, resulting in a significant imbalance between generators' mechanical power and electrical power between 4.0 s to 4.1 s, as shown in Fig. 15(b). This imbalance forces the rotor speed to drop until load shedding happens. The frequency transient after load shedding is caused by the dynamic response of the generators' governors. Note that in this case all grid-following inverters have similar dynamic behaviors because they have the same controller parameters.

Fig. 16(a)–(d) show the three-phase voltages of all the nodes at the primary voltage side, and the output reactive power of generators and inverters. It is seen that the voltage profile of Phase B is apparently "thicker" than those of Phases A and C, indicating the simulation was conducted on a three-phase unbalanced distribution system. The three-phase node voltages range from 0.948 pu to 0.983 pu before the event, and from 0.923 pu to 0.981 pu after the event. During load transients, the voltages drop to 0.750 pu and jump to 1.065 pu due to the response of the generators' exciters. The voltage profile is relatively poor because only the generators control the system voltage, resulting in a weak islanded system, and the tripping of Generator 1 further reduces the stiffness of the system. The grid-following inverters are controlled as current sources, so they do not control the voltage.

B. Loss of One Generator With All Grid-Forming Inverters

Figs. 17 and 18 show the simulation results for the same loss of generation event but with all inverters being grid-forming controlled. As shown in Fig. 17(a), the frequency nadir is significantly improved compared to the results for Case A, and therefore under-frequency load shedding is avoided. In Fig. 17(a), the frequencies of generators refer to their speeds, and the frequencies of inverters are obtained from their controllers (as shown in Fig. 9(b)). The improvement in the frequency nadir is caused by the grid-forming inverters quickly increasing their output power after the event because of their voltage source characteristics, thereby resulting in a much smaller imbalance between the mechanical power and electrical power of generators, as shown in Fig. 17(b).

As shown in Fig. 17(a), the frequencies are different between 4.0 s to 4.2 s and can be divided into four groups. The initial frequency drop of Generators 2 and 3 is decided by their inertia time constants H, and the frequency drop of inverters is decided by

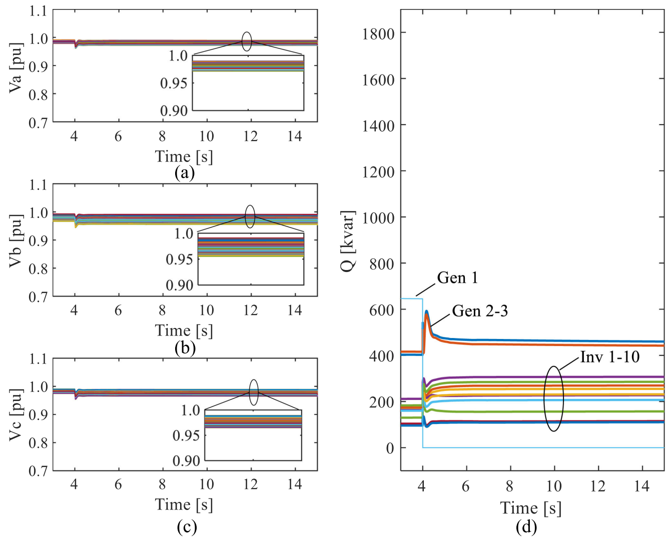

Fig. 18. Case B, grid-forming: (a)–(c) voltages and (d) reactive power.

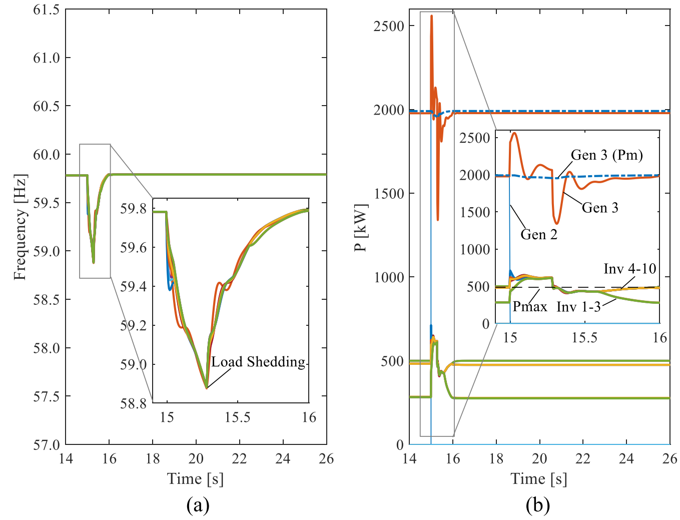

Fig. 19. Case C, grid-forming: (a) frequency and (b) active power.

TABLE I
COMPUTATION TIME SUMMARY

Case	Simulation Time (s)	Time Step (ms)	Computation Time (s)
A	15	2	2,014
B	15	5	801

their controllers. As shown in Fig. 17(b), because Inverters 9 and 10 already output their maximum power before the event, their overload mitigation controllers are activated after the tripping of Generator 1, so their frequencies drop faster than the other inverters, as shown in Fig. 17(a). In contrast, the output power of Inverters 1–3 do not exceed their maximum outputs during the load transient, so their frequencies are only governed by the *P-f* droop controllers and are the highest among all inverters. The output power of Inverters 4–8 exceeds their maximum power outputs between 4.07 s to 4.17 s, as shown in Fig. 17(b), so their frequencies are between those of the other two groups of inverters. All frequencies reach the same value in the steady state. The frequency drop in the steady state is decided by the droop slope of the generators' governors and the inverters' *P-f* droop controllers, which is set at 1% for both in this study.

Fig. 18(a)–(d) show the three-phase voltages of all the nodes at the primary voltage side, and the output reactive power of generators and inverters. The three-phase node voltages range from 0.966 pu to 0.992 pu before the event, and from 0.953 pu to 0.991 pu after the event. The voltage drops to 0.944 pu during load transients. Compared to Case A, the voltage profile is significantly improved, especially during load transients. This is because the internal voltages of grid-forming inverters remain constant during the event and they autonomously output necessary reactive power to maintain the system voltage, as shown in Fig. 18(d).

C. Loss of the Second Generator Following Case B

In case C, Generator 2 is tripped at 15 s after Case B occurs. The loss of two generators results in insufficient generation of the entire islanded system (7 MW generation versus 8 MW load). Therefore, both generators and inverters are overloaded, and the overload mitigation controllers of grid-forming inverters are activated to reduce the system frequency, thereby triggering under-frequency load shedding, as shown in Fig. 19(a) and (b).

The results in Fig. 19(a) and (b) are similar to the load shedding results for CERTS/AEP microgrid shown in Fig. 13, but the simulation is conducted in a large-scale, three-phase unbalanced distribution system that has thousands of nodes. Note that the under-frequency load shedding in Case C is different from that in Case A. In Case A, the inverters still have sufficient generation to supply all the loads, but the grid-following control cannot autonomously increase the output power of inverters to meet the load change; therefore, load shedding is triggered by the frequency transient of synchronous generators. In contrast, the under-frequency load shedding in Case C is caused by the overload situation of the entire islanded distribution system, in which some loads have to be tripped to guarantee the survival of the system.

The above simulation results show that with the developed inverter models, it is possible to examine the dynamic stability of large-scale, three-phase unbalanced distribution systems with high penetration of inverter-based DERs. Furthermore, results show that compared to traditional grid-following inverters, grid-forming inverters can significantly improve the voltage and frequency stability of islanded distribution systems. This finding at the distribution level is in accordance with findings at the transmission level [8], [9], [12], [13], [21].

D. Computation Time Summary

Table I summarizes the computation times needed for simulating Case A and Case B. The computation time for Case C is not listed as it is an event that occurs after Case B. Simulation was conducted on a Core I7-9850H 2.60 GHz computer that had 16 GB RAM, and the operation system was Windows 10

Enterprise. The simulation time was 15 s for each case. As seen in Table I, the computation time for Case A is significantly longer than that for Case B. This is because the authors found that the developed grid-following inverter model can only run at a relatively small simulation time step (e.g., 2 ms for Case A), due to the existence of a fast response inner current control loop. A larger simulation time step could cause numerical stability issues. In contrast, the developed grid-forming inverter model can run at a larger simulation time step without causing numerical stability issues (e.g., 5 ms for Case B). Note that the simulation efficiency could be further improved by optimizing the power flow solver, but doing so is beyond the scope of this paper.

VIII. Conclusion

Historically, distribution system planning studies have mainly focused on steady state and quasi-steady state analysis, and limited attention was paid to dynamic analysis. This paper develops three-phase, electromechanical models for both grid-following and grid-forming inverters, and integrates them into an open source, three-phase distribution network solver, thereby enabling dynamic simulation of large-scale, three-phase unbalanced distribution systems that feature high penetration of inverter-based DERs. The inverter models faithfully represent the control strategies used by typical grid-following and grid-forming inverters. The proposed inverter models are validated against electromagnetic simulation and field test data from the CERTS/AEP microgrid testbed, and studied in an islanded, three-phase unbalanced, 5252 node distribution feeder, which has multiple synchronous generator-based and inverter-based DERs. Simulation verifies the effectiveness of the developed inverter models for large-scale distribution system analysis. Study results show that compared to traditional grid-following inverters, the high penetration of grid-forming inverters can improve the voltage and frequency stability of islanded distribution systems.

Appendix

TABLE II
Grid-Following Controller Parameters

k_{pc}	k_{ic}	k	k_{pPLL}	k_{iPLL}
0.05 pu	5 pu	0.5 pu	50 pu	1,000 pu/s

TABLE III
Grid-Forming Controller Parameters

m_p	m_q	T	ω_0	k_{pv}
3.77 rad/s	0.05 pu	0.01 s	376.99 rad/s	0 pu
k_{iv}	k_{ppmax}	k_{ipmax}	P_{max}	P_{min}
5.86 pu/s	3 pu	60 pu/s	1 pu	0 pu

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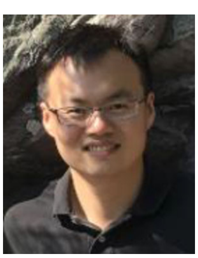

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